

LEGAL AID CHICAGO



Outcome Tagging Approach

- Summarize entire CAR dataset by Contact Session ID, grabbing the last value for each field:
 - Activity Name
 - EP Name
 - Flow Name
 - Agent Name
 - + Time difference between last activity and final row in ID
- Convert case-when to Python dictionary logic to tag outcomes/subtypes using spreadsheet
 - Meeting conditions for last activity (and other fields when indicated in comments column), mapping them to their respective outcome/subtype through the dictionary
- Add capability to traverse back to preceding menus/activities for special cases (i.e., courtesy callback)

```
If EP name = 'Closed Queue Menu Telephony EP',  
then Negative, and type = "Closed queue",  
otherwise Positive, type = 'Reached voicemail', if  
EP name = 'Closed Hours-Holidays Menu  
Telephony EP' then neutral, and type = 'After hours  
call'
```

Pseudocode

```
When  
- Last activity = ClinicVoicemailTransfer AND EP name =  
  "Closed Queue Menu Telephony EP"  
  - THEN Negative, Closed Queue  
- Last activity = ClinicVoicemailTransfer AND EP name !=  
  "Closed Queue Menu Telephony EP"  
  - THEN Positive, Voicemail Reached  
- Last activity = ClinicVoicemailTransfer AND EP name =  
  'Closed Hours-Holiday Menu Telephony EP'  
  - THEN Neutral, After Hours Call
```

Overview of Outcome Classifications

Positive

Legal Aid Chicago is able to effectively address the caller's issue by:

- Speaking to an agent during the initial call or through a courtesy callback
- Receiving information about relevant resources through a recorded message

59,237 Calls - 24%

Neutral

The caller's issue was not addressed, but should not be addressed by Legal Aid Chicago

- The call was received during closed hours
- Legal Aid Chicago doesn't serve their issue

57,421 Calls - 23%

Negative

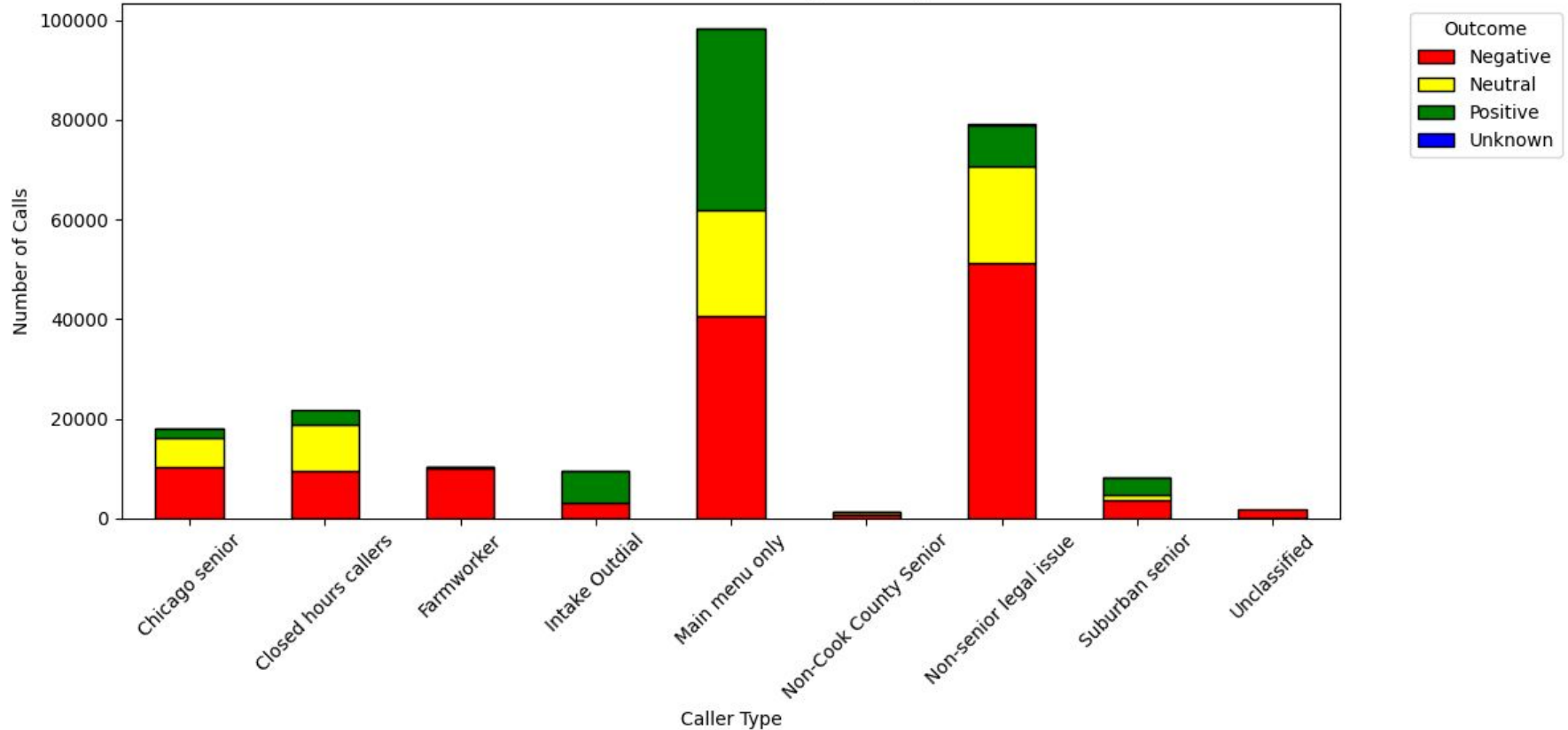
The caller's issue was not addressed by Legal Aid Chicago due to:

- The call being abandoned
- The call reaching a closed queue
- Courtesy callback is unsuccessful

131,278 Calls - 53%

Overview of Call Outcomes

Distribution of Outcomes by Caller Type



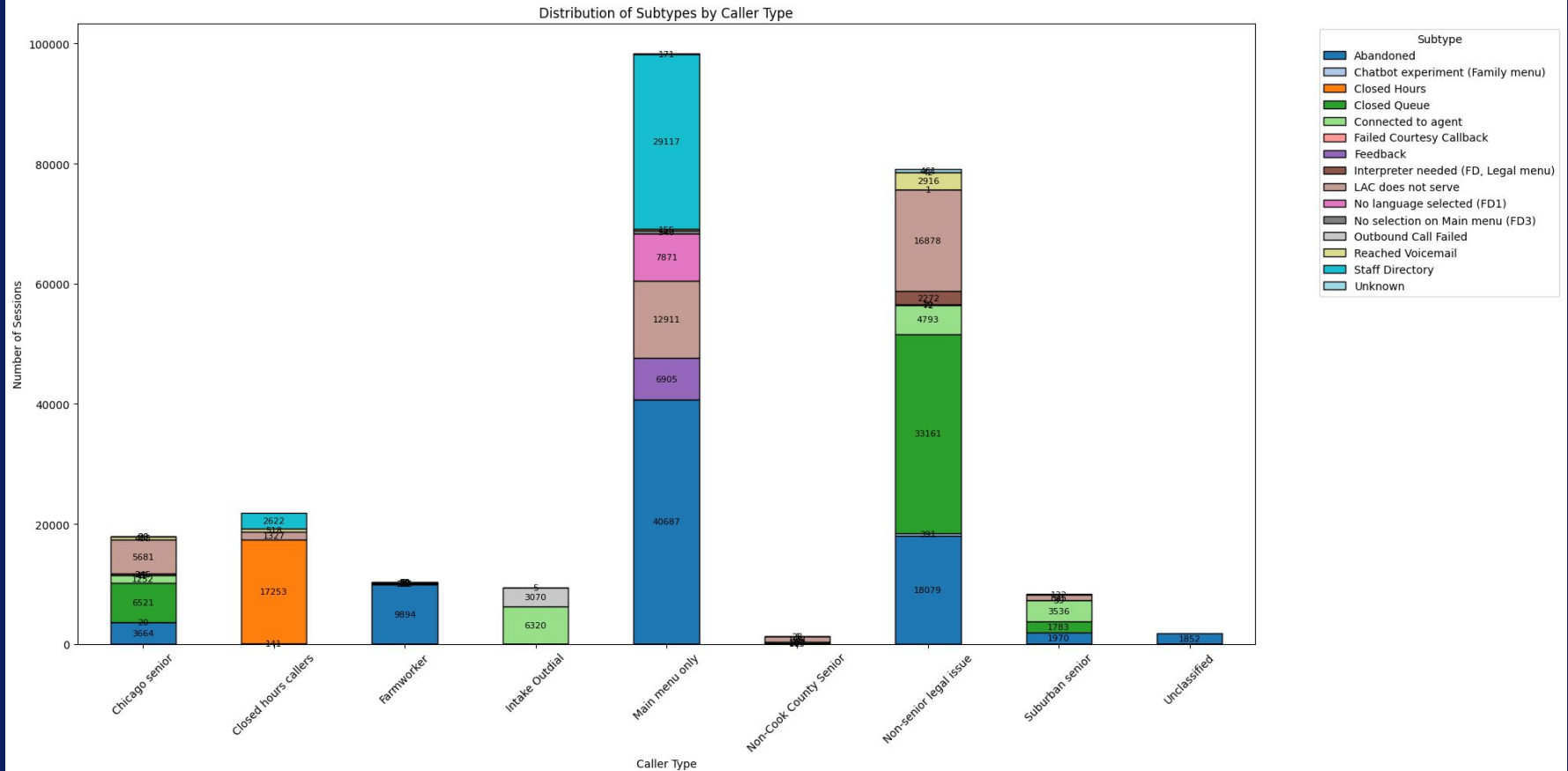
Call Breakdown by Call Type

Call type	Call sub-type	Proportion	Total calls	Calls per weekday
Main Menu		40%	98,357	266
Legal Issue Calls		44%	108,519	293
	Non-Seniors	32%	79,085	214
	Chicago Seniors	7%	17,975	49
	Suburban Seniors	3%	8,305	22
	Non-Cook Seniors	1%	1,302	4
	Unclassified Callers	1%	1,852	5
Other Calls		17%	41,566	112
	Closed Hours	9%	21,861	59
	Farmworker Menu	4%	10,310	28
	Intake Outdial	4%	9,395	25
Total		100%	248,442	668

Call Outcomes by Call Type

Call type	Call sub-type	Positive	Proportion	Neutral	Proportion	Negative	Proportion
Main Menu		36,177	36%	21,322	22%	40,687	41%
Legal Issue Calls		13,499	12%	26,787	25%	67,905	63%
	Non-Seniors	8,161	10%	19,445	25%	51,312	65%
	Chicago Seniors	1,752	10%	5,995	33%	10,201	57%
	Suburban Seniors	3,536	43%	845	10%	3,792	46%
	Non-Cook Seniors	50	4%	502	39%	748	58%
	Unclassified Callers	0	0%	0	0%	1,852	100%
Other Calls		9,561	23%	9,312	22%	22,686	55%
	Closed Hours	3140	14%	9,208	42%	9,513	44%
	Farmworker Menu	101	1%	104	1%	10,103	98%
	Intake Outdial	6320	67%	0	0%	3,070	32%
Total		59,237	24%	57,421	23%	131,278	53%

Overview of Call Subtypes



Call Breakdown by Legal Issue

Call type	Call sub-type	Total calls	Proportion
Legal Issue Calls (Non-Senior Menu)			
	Legal Menu	22,469	28.4%
	Family Menu	20,762	26.2%
	Housing Menu	12,769	16.1%
	Other Legal Menu	9,175	11.6%
	Benefits Menu	5,111	6.4%
	Employment Menu	3,997	5.1%
	Immigration Menu	3,268	4.1%
	HIV Menu	1,392	1.8%
Total		79,085	100%

Legal Menu Outcomes

Outcome	Subtype	Count	Proportion
Positive		2,718	12.1%
	Reached Voicemail	2001	8.9%
	Connected to Agent	717	3.2%
Neutral		3025	13.5%
	Interpreter needed (FD)	2272	15.4%
	LAC does not serve	746	3.3%
	Unknown	7	0.03%
Negative		16,270	74.4%
	Abandoned	11,435	50.9%
	Closed Queue	5,273	23.5%
	Failed Courtesy Callback	12	0.05%
Unknown		6	0.03%
Total		22,469	100%

Family Menu Outcomes

Outcome	Subtype	Count	Proportion (of Family Menu Outcomes)
Positive		2,195	10.6%
	Connected to Agent	1,804	8.7%
	Chatbot Experiment	391	1.9%
Neutral		3,192	15.4%
	LAC does not serve	3,912	15.4%
Negative		15,281	73.5%
	Closed Queue	13,217	63.7%
	Abandoned	2,037	9.8%
	Failed Courtesy Callback	27	0.1%
Unknown		94	0.5%
Total		20,762	100%

Housing Menu Outcomes

Outcome	Subtype	Count	Proportion (of Housing Menu Outcomes)
Positive		1,115	8.7%
	Connected to Agent	1,115	8.7%
Neutral		2,556	20.0%
	LAC does not serve	2,556	20.0%
Negative		9,070	71.0%
	Closed Queue	6,283	69.3%
	Abandoned	2,770	21.7%
	Failed Courtesy Callback	17	0.1%
Unknown		28	0.2%
Total		12,769	100%

Benefits Menu Outcomes

Outcome	Subtype	Count	Proportion (of Benefits Menu Outcomes)
Positive		621	12.1%
	Connected to Agent	516	10.1%
	Reached Voicemail	105	2.1%
Neutral		468	9.2%
	LAC does not serve	468	9.2%
Negative		4,005	78.4%
	Closed Queue	3,656	71.5%
	Abandoned	341	8.5%
	Failed Courtesy Callback	8	0.2%
Unknown		17	0.3%
Total		5,111	100%

Employment Menu Outcomes

Outcome	Subtype	Count	Proportion (of Benefits Menu Outcomes)
Positive		474	11.9%
	Connected to Agent	474	11.9%
Neutral		449	11.2%
	LAC does not serve	449	11.2%
Negative		3,058	76.5%
	Closed Queue	2,832	70.9%
	Abandoned	222	5.6%
	Failed Courtesy Callback	4	0.1%
Unknown		16	0.4%
Total		3,997	100%

Issues

For Krish,

- There seems to be calls where the caller traverses through the Legal Menu, but ultimately makes a selection on Main Menu - these are currently considered 'Non-Senior Legal Issues'
 - Ex: Caller goes through Housing and Benefits Menus before selecting staff directory transfer from the Main Menu (see appendix)
 - Similar issue for Pre-Legal Seniors, but it seems to include calls where the last menu was Seniors Menu
 - Seems to be relatively few calls: Main Menu (136), Pre-Legal Seniors (5)
 - Unsure of how adding outcomes would have impacted this tagging
 - Not included in the call breakdown by legal issue
- All outcomes for Other Legal Menu are neutral and this fit with examples examined, but may require further verification to ensure that this holds for all cases
 - Opted to exclude from drill down of five most popular menus as it seemed to not be illustrative
- Currently do not have a distinction between Legal Menu 1 and 2

Takeaways & Recommendations

Takeaways:

- Negative outcomes represent 71-78% of outcomes for the most popular legal menus, and these tend to largely be due to closed queues
- Abandonment was the most common negative outcome for Legal Menus 1 & 2 → could reflect that callers stay on or return to this menu if they're unable to find their specific legal issue, or they're overwhelmed by the options

Recommendations:

1. Allowing the option to schedule a courtesy callback even when there is a closed queue
 - a. So people don't have to call multiple times, reach a closed queue, and abandon their call
 - b. Helps a majority of non-senior legal issues clients
 - i. Closed queue contributes to the greatest number of negative outcomes for the popular legal menus



Thank you!

Appendix

Call Breakdown by Call Type - Unknown

Call type	Call sub-type	Unknown
Main Menu		171
Legal Issue Calls		328
	Non-Seniors	167
	Chicago Seniors	27
	Suburban Seniors	132
	Non-Cook Seniors	2
	Unclassified Callers	0
Other Calls		7
	Closed Hours	0
	Farmworker Menu	2
	Intake Outdial	5
Total		

Outcome Tagging Code

```
# Function to assign outcomes
def assign_outcomes(df, activity_col='Final Activity', ep_col='EP Name', outcome_col='Outcome', subtype_col='Subtype', rules_outcome=None,
                    rules_subtype=None, conditions=None, cond_outcomes=None, cond_subtypes=None, default_outcome='Unknown', default_subtype='Unknown'):

# Adding outcome & subtype for activities with multiple options
    if conditions and cond_outcomes and cond_subtypes:
        df[outcome_col] = np.select(conditions, cond_outcomes, default=default_outcome)
        df[subtype_col] = np.select([conditions, cond_subtypes], default=default_subtype)]
    else:
        df[outcome_col] = default_outcome
        df[subtype_col] = default_subtype

# Adding outcome & subtype for activities with a single options
    if rules_outcome:
        mask_outcome = df[activity_col].isin(rules_outcome.keys())
        df.loc[mask_outcome, outcome_col] = df.loc[mask_outcome, activity_col].map(rules_outcome)

    if rules_subtype:
        mask_subtype = df[activity_col].isin(rules_subtype.keys())
        df.loc[mask_subtype, subtype_col] = df.loc[mask_subtype, activity_col].map(rules_subtype)

    return df

assign_outcomes(all_act, rules_outcome=outcome_rules,
                rules_subtype=subtype_rules,
                conditions=conditions,
                cond_outcomes=outcome_choices,
                cond_subtypes=subtype_choices,
                default_outcome='Unknown',
                default_subtype='Unknown')
```

Outcome Tagging Code

```
outcome_rules = {
    'ClosedQueueMenu' : 'Negative',
    'StaffDirectoryEnglishTransfer': 'Positive',
    'LanguageSelectionMenu': 'Negative',
    'ClinicVoiceMailTransfer': 'Positive',
    'MainMenu': 'Negative',
    'DisconnectContact1': 'Negative',
    'LegalMenu2': 'Negative',
    'FarmworkerMainMenu': 'Negative',
    'SetCallerID': 'Positive',
    'LegalServerScreenPop': 'Positive',
    'FrontDeskTransfer1': 'Neutral',
    'ClosedMenu': 'Neutral',
```

```
# Relevant menus for checking OtherLegalMenu outcome
menus_of_interest = [
    'OtherLegalPersonalInjuryMenu',
    'OtherLegalCriminalCaseMenu',
    'OtherLegalOtherMenu',
    'ClinicVoicemailTransfer'
]

# For each contact session ID, checking if activity names contain menus for OtherLegalMenu rule
all_act['Touched_Legal_Menu'] = [
    all_act.groupby('Contact Session ID')['Activity Name']
        .transform(lambda x: x.isin(menus_of_interest).any())
]

# Conditions for each outcome
cond1 = (all_act['Final Activity'] == 'ClinicVoiceMailTransfer') & \
        (all_act['EP Name'] == 'Closed Queue Menu Telephony EP')
cond2 = (all_act['Final Activity'] == 'ClinicVoiceMailTransfer') & \
        (all_act['EP Name'] == 'Closed Hours-Holidays Menu Telephony EP')
cond3 = (all_act['Final Activity'] == 'DisconnectContact1') & \
        (all_act['EP Name'] == 'Closed Hours-Holidays Menu Telephony EP')
cond4 = (all_act['Final Activity'] == 'DisconnectContact1') & \
        (all_act['EP Name'] == 'Closed Queue Menu Telephony EP')
cond5 = (all_act['Final Activity'] == 'DisconnectContact1') & \
        (all_act['EP Name'] == 'Courtesy Callback Telephony EP')
cond6 = (all_act['Final Activity'] == 'DisconnectContact1') & \
        (all_act['EP Name'] == 'Pre-Legal Menu Seniors Menu Telephony EP')
cond7 = all_act['Touched_Legal_Menu']

conditions = [cond1, cond2, cond3, cond4, cond5, cond6, cond7]
outcome_choices = ['Negative', 'Neutral', 'Neutral', 'Negative', 'Negative', 'Neutral', 'Neutral']
subtype_choices = ['Closed Queue', 'Closed Hours', 'Closed Hours', 'Closed Queue',
                    'Failed Courtesy Callback', 'LAC does not serve', 'LAC does not serve']
```

Update Final Menu Code

```
# EPs that should not be used as final menu
ignored_eps = ['All LAC Queues Telephony EP', 'Courtesy Callback Telephony EP', 'Closed Queue Menu Telephony EP']

# Function to
def update_final_menu(df, final_menu_col="Final Menu", ep_col="EP Name", id_col="Contact Session ID", time_col="Activity Start Timestamp", ignored_eps=None):

    # Making sure rows for a contact session ID are ordered by time
    df = df.sort_values([id_col, time_col])

    # Filter to keep valid EPs
    valid_eps = df[~df[ep_col].isin(ignored_eps)]

    # Find last valid EP per contact session
    last_ep = valid_eps.groupby(id_col)[ep_col].last()

    # Update Final Menu column for all rows using map
    df[final_menu_col] = df[id_col].map(last_ep)

    return df

final_act2 = update_final_menu(final_act, ignored_eps=ignored_eps)
```

✓ 2.6s

Sub-menu Outcomes

```
legal_menu = final_legal.loc[final_legal['Final Menu'] == 'Legal Menu Telephony EP', :]
```

```
summary = legal_menu.groupby('Outcome').size().reset_index(name='Count').sort_values(by='Count', ascending= False)  
summary
```

✓ 0.0s

	Outcome	Count
0	Negative	16720
1	Neutral	3025
2	Positive	2718
3	Unknown	6

```
summary = legal_menu.groupby(['Outcome', 'Subtype']).size().reset_index(name='Count').sort_values(by='Count', ascending= False)  
summary
```

✓ 0.0s

	Outcome	Subtype	Count
0	Negative	Abandoned	11435
1	Negative	Closed Queue	5273
3	Neutral	Interpreter needed (FD, Legal menu)	2272
7	Positive	Reached Voicemail	2001
4	Neutral	LAC does not serve	746
6	Positive	Connected to agent	717
2	Negative	Failed Courtesy Callback	12
5	Neutral	Unknown	7
8	Unknown	Unknown	6

Example of Main Menu - Non-Senior Legal Issue

Contact Session ID	EP Name	Flow Name	Activity Name	Activity Start Timestamp	Queue Name	Agent Name	Termination Reason	hour	caller_type	Overall_tag	Final Activity	Outcome	Subtype	Final Menu
815	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Legal Menu Telephony EP	NaN	LegalMenu2	2025-07-29 10:57:44	NaN	NaN	NaN	10	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
816	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 NaN	LegalBenefitsMenu	NaN	2025-07-29 11:01:07	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
817	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Legal Benefits Menu Telephony EP	NaN	BenefitsMenu	2025-07-29 11:01:07	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
818	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Legal Benefits Menu Telephony EP	NaN	BenefitsMenu	2025-07-29 11:01:56	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
819	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Legal Benefits Menu Telephony EP	NaN	BenefitsMenu	2025-07-29 11:02:45	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
320	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 NaN	LegalMenu	NaN	2025-07-29 11:03:13	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
821	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Legal Menu Telephony EP	NaN	LegalMenu2	2025-07-29 11:03:13	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
322	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Legal Menu Telephony EP	NaN	LegalMenu2	2025-07-29 11:03:14	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
323	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Legal Menu Telephony EP	NaN	LegalMenu2	2025-07-29 11:03:24	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
324	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 NaN	LACMain	NaN	2025-07-29 11:03:26	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
325	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Main Number Telephony EP	NaN	MainMenu	2025-07-29 11:03:26	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP
326	f4fa1a26-3b4a-43b8-9c51-358b8dbb6321 Main Number Telephony EP	NaN	StaffDirectoryEnglishTransfer	2025-07-29 11:03:27	NaN	NaN	NaN	11	Non-senior legal issue	Legal issue	StaffDirectoryEnglishTransfer	Positive	Staff Directory	Main Number Telephony EP